

Consequences of Rybin’s Counterexample: Convex-Decomposition Failure and a Planar $9/8$ Lower Bound for Cost-Preserving Unsplittable Flow

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Abstract

Rybin recently disproved Goemans’ cost-preserving strengthening of the Dinitz–Garg–Goemans theorem with a seven-vertex planar acyclic instance whose fractional flow costs 58 while every unsplittable flow within the additive max-demand bound costs at least 60 [5]. We record two consequences. First, the same instance refutes the cost-and-two-sided-bounds conjecture and the convex-decomposition strengthening of Morell and Skutella; their costless two-sided existence conjecture is untouched. Second, a parametric form of the instance shows that every universal cost-preserving rounding theorem on planar acyclic digraphs requires additive congestion at least $(9/8 - o(1))D$, where D is the maximum demand. Combined with the planar factor-two theorem of Traub, Vargas Koch, and Zenklusen, the optimal planar cost-preserving constant lies in $[9/8, 2]$. The underlying mechanism is a violated triangle stable-set facet, which we state as a reusable design principle.

1 Introduction

The theorem of Dinitz, Garg, and Goemans states that a feasible fractional single-source flow can always be rounded to an unsplittable flow while increasing every arc load by at most the maximum demand [2]. Goemans proposed that the same additive bound should remain achievable with no increase in cost; the conjecture was recorded in the minimum-cost literature by Skutella [6] and guided work on min-cost unsplittable flow for roughly a quarter-century [3, 4, 7].

In July 2026 Dmitry Rybin disproved the conjecture with an explicit seven-vertex planar instance, constructed in a public AI-assisted research session [5]. Section 3 reproduces that instance and its certificate, with full credit to Rybin, because the two contributions of this note are read off directly from it: the corollaries in Section 4 and the quantitative planar lower bound in Section 5.

2 Preliminaries

Let $G = (V, A)$ be a directed acyclic graph with common source s , terminals $t_1, \dots, t_k$, and demands $d_1, \dots, d_k > 0$. A fractional flow $x \in \mathbb{R}_{\geq 0}^A$ satisfies the corresponding divergence constraints. An unsplittable flow chooses one directed s – t_i path P_i per terminal and has load

$$y_a = \sum_{i: a \in P_i} d_i \quad (a \in A).$$

Write $D := \max_i d_i$. The Dinitz–Garg–Goemans theorem guarantees an unsplittable y with $y_a \leq x_a + D$ for all arcs [2]. Goemans’ cost conjecture asked, for every nonnegative arc-cost vector c , for such a y satisfying additionally $c^\top y \leq c^\top x$.

3 Rybin's instance

Everything in this section is due to Rybin [5]; we reproduce it to keep the note self-contained. Let

$$V = \{s, u, v, w, t_1, t_2, t_3\}, \quad (d_1, d_2, d_3) = (15, 10, 15), \quad D = 15,$$

with arcs, fractional loads, and costs as in Table 1.

Table 1: Rybin's flow instance [5].

arc a	x_a	c_a
$s \rightarrow t_1$	10	2
$s \rightarrow t_2$	6	3
$s \rightarrow u$	24	0
$u \rightarrow t_3$	10	2
$u \rightarrow v$	14	0
$v \rightarrow t_1$	5	0
$v \rightarrow w$	9	0
$w \rightarrow t_2$	4	0
$w \rightarrow t_3$	5	0

Each terminal has exactly two source-terminal paths: the expensive direct options $E_1 = st_1$, $E_2 = st_2$, $E_3 = sut_3$ (each of total integral cost 30 after multiplication by its demand) and the zero-cost options $Z_1 = suvt_1$, $Z_2 = suvwt_2$, $Z_3 = suvwt_3$. The fractional flow is feasible with cost $c^\top x = 58$, and a routing satisfies $y \leq x + D\mathbf{1}$ if and only if it uses at most one of Z_1, Z_2, Z_3 : each pair of zero-cost choices overloads one of the arcs su, uv, vw by exactly one unit beyond $x_a + D$. Hence every capacity-good routing uses at least two expensive paths and costs at least 60, and (E_1, E_2, Z_3) attains 60:

$$\min\{c^\top y : y \text{ unsplittable, } y \leq x + D\mathbf{1}\} = 60 > 58 = c^\top x.$$

4 Two conjectures of Morell and Skutella fall with it

Morell and Skutella conjectured that every fractional single-source flow is a convex combination of unsplittable flows satisfying the simultaneous bounds $x_a - D \leq y_a \leq x_a + D$ on every arc, and noted that such a decomposition would provide no-more-expensive members for arbitrary arc costs [4, Conjecture 2].

Corollary 4.1. *Rybin's instance refutes both the cost-and-two-sided-bounds conjecture and the convex-decomposition strengthening of Morell and Skutella. It does not refute their costless two-sided existence conjecture.*

Proof. Every two-sided-good routing is in particular upper-good, and every upper-good routing costs at least 60. If $x = \sum_r \lambda_r y^{(r)}$ were a convex combination of two-sided-good unsplittable flows, then

$$58 = c^\top x = \sum_r \lambda_r c^\top y^{(r)} \geq 60,$$

a contradiction. The same separator applies verbatim to the cost-enhanced two-sided statement. The costless existence statement survives: the instance admits two-sided-good routings, so no contradiction arises there. $\square$

Remark 4.2. The underlying undirected graph is a subdivision of K_4 , the minimal forbidden topology for series-parallel graphs, where the convex-decomposition statement is a theorem [1]. The boundary is therefore sharp: series-parallel digraphs admit the decomposition, while a single K_4 subdivision already destroys it.

5 A parametric family and the planar $9/8$ bound

Normalize $D = 1$ and retain the topology of Table 1. Let

$$d_1 = d_3 = 1, \quad d_2 = b \in (0, 1],$$

let the cheap-path probabilities in the fractional path decomposition be $p_1 = p_3 = r$ and $p_2 = q$, and give each expensive path total integral cost one: $c_{st_1} = 1$, $c_{st_2} = 1/b$, $c_{ut_3} = 1$, all other costs zero. The fractional cost is then $3 - (2r + q)$.

Proposition 5.1. *These parameters define a strict counterexample to the additive- D cost conjecture whenever*

$$2r + q > 1, \quad b(1 - q) > r, \quad 2r + bq < 1.$$

More generally, every no-more-expensive routing has upper deviation at least

$$\min\{1 + b(1 - q) - r, 2 - 2r - bq\}.$$

Proof. Since $2r + q > 1$, the fractional cost is strictly below 2. Integral routing costs are integers, so a no-more-expensive routing uses at most one expensive path, hence at least two cheap paths. The pair (Z_1, Z_2) exceeds the fractional load on su by $1 + b(1 - q) - r$; the pair (Z_1, Z_3) exceeds the load on uv by $2 - 2r - bq$; the pair (Z_2, Z_3) exceeds the load on vw by the same amount as (Z_1, Z_2) . The last two displayed inequalities make both expressions strictly larger than $D = 1$. $\square$

Define α_{planar} as the infimum of constants α such that every planar acyclic instance admits an unsplittable y with $c^\top y \leq c^\top x$ and $y_a \leq x_a + \alpha D$ for all arcs.

Theorem 5.2.

$$\frac{9}{8} \leq \alpha_{\text{planar}} \leq 2.$$

Moreover, $9/8$ is the supremum of the lower bounds generated by the symmetric family of Proposition 5.1.

Proof. The upper bound is the planar cost theorem of Traub, Vargas Koch, and Zenklusen [8]. For the lower bound take

$$b = \frac{3}{4}, \quad r = \frac{1}{4}, \quad q = \frac{1}{2} + \varepsilon.$$

The fractional cost is $2 - \varepsilon$, so every no-more-expensive routing uses two cheap paths, and each of the three cheap pairs exceeds its corresponding fractional arc load by $\frac{9}{8} - \frac{3}{4}\varepsilon$. Letting $\varepsilon \downarrow 0$ proves $\alpha_{\text{planar}} \geq 9/8$.

For optimality within the symmetric family, decrease q to the boundary $q = 1 - 2r$; this can only increase both expressions in Proposition 5.1. If $b \geq 1/2$, the two affine expressions balance at $r = 1 - b$, where their common value is $b(3 - 2b)$, maximized at $b = 3/4$ with value $9/8$. If $b < 1/2$, neither expression exceeds 1. Strict cost separation requires $q > 1 - 2r$, so $9/8$ is approached but not attained. $\square$

6 The triangle-facet mechanism

Let $z_i = 1$ indicate that terminal i uses its cheap path Z_i . Capacity-good integral routings satisfy the triangle stable-set facet $z_1 + z_2 + z_3 \leq 1$, while the fractional cheap-path marginals of Rybin’s instance are $(\frac{1}{3}, \frac{2}{5}, \frac{1}{3})$ with sum $\frac{16}{15} > 1$. The cost vector is a nonnegative separator for this violated facet, which is why the same instance simultaneously kills the cost conjecture and the convex decomposition. As a design principle: identify binary local decisions, characterize the capacity-feasible integral patterns as stable sets of a conflict graph, place the fractional marginals outside the stable-set polytope, and read the cost vector off the violated facet.

7 Verification and disclosure

The accompanying repository contains Rybin’s instance in machine-readable form, an exhaustive exact verifier over all eight routings (`verify_dgg.py`), and an enumerator for the parametric family (`search_dgg_family.py`) confirming that $9/8$ is the supremum over small-denominator parameters. A generative AI system (GPT-5.6 Pro) assisted with exploration, checking, code drafting, and exposition during the research session in which these observations were made; it is not an author. The author assumes responsibility for verification, attribution, and the final text. The base counterexample is Rybin’s; this note claims only the corollaries, the parametric analysis, and the exposition.

References

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